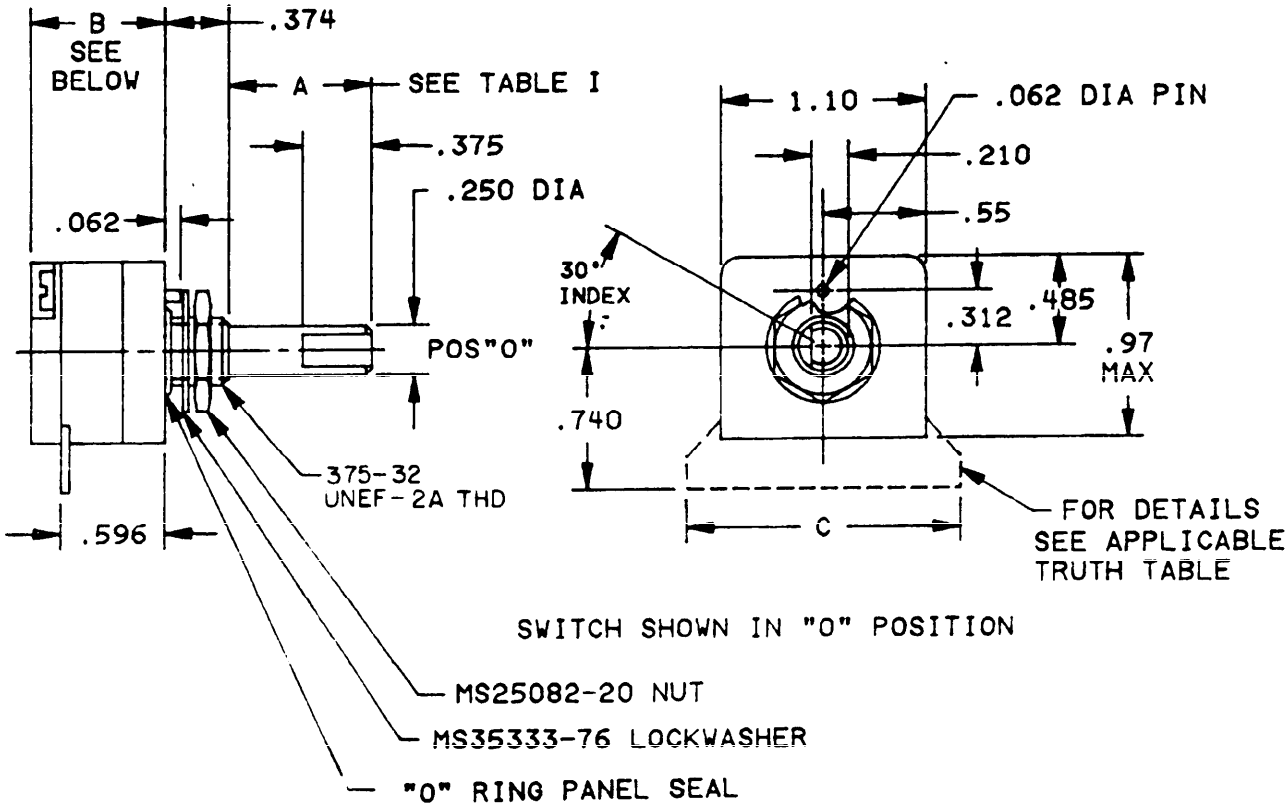
MILITARY SPECIFICATION SHEET

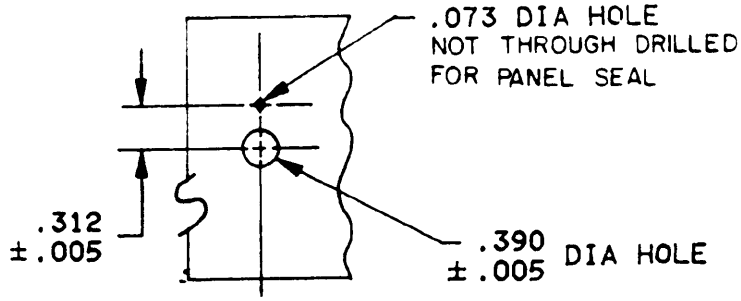
SWITCHES, ROTARY PRINTED CIRCUIT DECK(S), 125 MILLIAMPERES AND LOW LEVEL, SEALED, RFI SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-3786.



TOTAL NO. OF DECKS	DIMENSION B ±.03
1	.75
2	1.12
3	1.47
4	1.82
5	2.17
6	2.52



PANEL CUTOUT

FIGURE 1. Dimensions and configurations.

FSC 5930

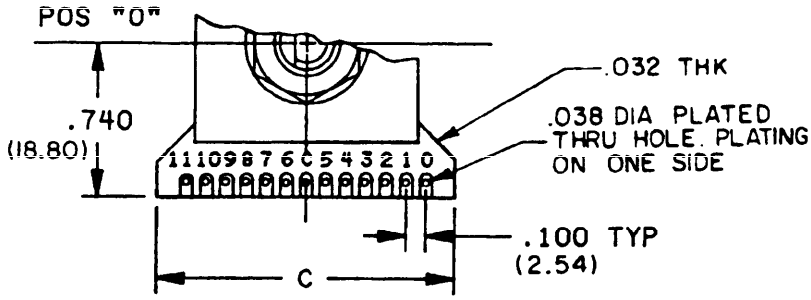
Source: https://assist.dla.mil -- Downloaded: 2026-10-16T21:42Z
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INCHES	MM	INCHES	MM	INCHES	MM
.005	.13	.374	9.50	.75	19.05
.03	.76	.375	9.53	.97	24.64
.062	1.57	.390	9.91	1.10	27.94
.073	1.85	.435	11.05	1.12	28.45
.210	5.33	.55	13.97	1.47	37.34
.250	6.35	.596	15.14	1.82	46.23
.312	7.92	.740	18.80	2.17	55.12
				2.52	64.01

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.76 mm) for two place decimals.

FIGURE 1. Dimensions and configurations - Continued.

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TERMINAL LAYOUT

TRUTH TABLE 1

SHAFT POS	COMMON "C" CONN. TO:											
	0	1	2	3	4	5	6	7	8	9	10	11
0	●											
1		●										
2			●									
3				●								
4					●							
5						●						
6							●					
7								●				
8									●			
9										●		
10											●	
11												●

TERMINAL LAYOUT

NO. OF POSITIONS	DIMENSION C
8	1.100 (30.73)
10	1.330 (33.78)
12	1.560 (39.62)

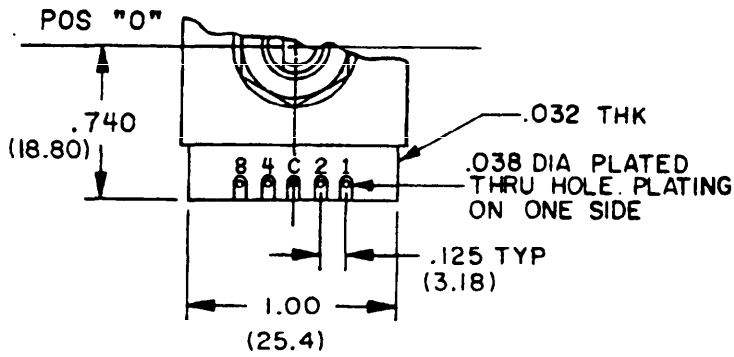
NOTES:
1. 8 position uses positions 0 thru 7.
2. 10 position uses positions 0 thru 9.
3. Terminals 8, 9, 10, and 11 are not provided on 8 position switch.
4. Terminals 10 and 11 are not provided on 10 position switch.

FIGURE 2. Truth tables and terminal configurations.

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TRUTH TABLE 2

SHAFT POS.	COMMON "C" CONN. TO:			
	1	2	4	8
0				
1	●			
2		●		
3	●	●		
4			●	
5	●		●	
6		●	●	
7	●	●	●	
8				●
9	●			●
10		●		●
11	●	●		●



TERMINAL LAYOUT

NOTES:
1. 8 position uses shaft position 0 thru 7.
2. 10 position uses shaft position 0 thru 9.
3. Terminal 8 is not provided on 8 position switch.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Millimeters are in parentheses.

FIGURE 2. Truth tables and terminal configurations - Continued.

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TABLE I. Switch characteristics and part numbers.

Part number M3786/43-	Indexing (angle of throw)	Number of positions (no stops)	Truth table	Number of decks	Shaft length (dim. A)
001	45	8	1	1	.750
002	45	8	1	1	.500
003	45	8	1	2	.750
004	45	8	1	2	.500
005	45	8	1	3	.700
006	45	8	1	3	.500
007	45	8	1	4	.750
008	45	8	1	4	.500
009	45	8	1	5	.750
010	45	8	1	5	.500
011	45	8	1	6	.750
012	45	8	1	6	.500
013	45	8	2	1	.750
014	45	8	2	1	.500
015	45	8	2	2	.750
016	45	8	2	2	.500
017	45	8	2	3	.750
018	45	8	2	3	.500
019	45	8	2	4	.750
020	45	8	2	4	.500
021	45	8	2	5	.750
022	45	8	2	5	.500
023	45	8	2	6	.750
024	45	8	2	6	.500
025	36	10	1	1	.750
026	36	10	1	1	.500
027	36	10	1	2	.750
028	36	10	1	2	.500
029	36	10	1	3	.750
030	36	10	1	3	.500

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TABLE I. Switch characteristics and part numbers - Continued.

Part number M3786/43-	Indexing (angle of throw)	Number of positions (no stops)	Truth table	Number of decks	Shaft length (dim. A)
031	36	10	1	4	.750
032	36	10	1	4	.500
033	36	10	1	5	.750
034	36	10	1	5	.500
035	36	10	1	6	.750
036	36	10	1	6	.500
037	36	10	2	1	.750
038	36	10	2	1	.500
039	36	10	2	2	.750
040	36	10	2	2	.500
041	36	10	2	3	.750
042	36	10	2	3	.500
043	36	10	2	4	.750
044	36	10	2	4	.500
045	36	10	2	5	.750
046	36	10	2	5	.500
047	36	10	2	6	.750
048	36	10	2	6	.500
049	30	12	1	1	.750
050	30	12	1	1	.500
051	30	12	1	2	.750
052	30	12	1	2	.500
053	30	12	1	3	.750
054	30	12	1	3	.500
055	30	12	1	4	.750
056	30	12	1	4	.500
057	30	12	1	5	.750
058	30	12	1	5	.500
059	30	12	1	6	.750
060	30	12	1	6	.500

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TABLE I. Switch characteristics and part numbers - Continued.

Part number M3786/43-	Indexing (angle of throw)	Number of positions (no stops)	Truth table	Number of decks	Shaft length (dim. A)
061	30	12	2	1	.750
062	30	12	2	1	.500
063	30	12	2	2	.750
064	30	12	2	2	.500
065	30	12	2	3	.750
066	30	12	2	3	.500
067	30	12	2	4	.750
068	30	12	2	4	.500
069	30	12	2	5	.750
070	30	12	2	5	.500
071	30	12	2	6	.750
072	30	12	2	6	.500

REQUIREMENTS

Dimensions and configuration: See figures 1 and 2.

Construction: F.

Angles of throw: 45, 36, and 30 degrees.

Temperature-life characteristic: B, test loads in accordance with table II.

TABLE II. Electrical test loads.

Environmental conditions	Resistive load
At atmospheric pressure	125 mA @ 28 V dc
At reduced barometric pressure	125 mA @ 28 V dc
Life, low level switching	10 mA max @ 30 mV max (open circuit)

Rotational torque: 0.5 in-lb minimum, 1.88 in-lbs maximum (total number of decks 1-6).

Vibration: Grade 3 (15 G, 10-2000 Hz).

Shock: Type M (100 G, sawtooth 6 ms).

Altitude rating: C (70,000 feet).

Terminal design: (For soldered wire connection) .038 plated thru holes.

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Contact resistance:

Initial: 100 milliohms maximum.

Following life (rotational): 160 milliohms maximum (rated), 1 ohm maximum (low level).

Dielectric withstanding voltage:

At atmosphere pressure: 750 V rms.

At reduced barometric pressure: 250 V rms.

Application of test voltage: Between switch terminals and between terminals to case.

Moisture resistance: Applicable.

Sand and dust: Applicable.

Explosion: Applicable.

Positive shaft grounding: Applicable.

RFI shielding: Applicable.

Mounting hardware: Each switch shall be supplied with one hexagon nut (MS25082-20), one conductive "O" ring panel seal (MS28775-012) and one lock washer (MS35333-76).

Part number: See table I.

QUALIFICATION

Extent of qualification: 24 sample units of M3786/43-059 shall be tested per qualification inspection of MIL-S-3786. This qualifies M3786/43-001 through M3786/43-072.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, OS
Air Force - 17,99
DLA - ES

User activities:
Army - AT, AV, AR
Navy - MC, CG
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Air Force - 85

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